# Qiaoya Wu

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## Education

**University of Illinois Urbana-Champaign (UIUC)**,
Ph.D. in Astronomy — Aug 2021 – present

Expected graduation: May 2027

Advisor: Prof. Yue Shen

Research areas: Quasar spectral surveys; AGN accretion physics and photoionization

**Xiamen University (XMU)**, B.S. in Astronomy — Sep 2017 – Jun 2021

Advisors: Prof. Hao-Ran Yu and Prof. Jianfeng Wu

Research areas: Cosmological N-body simulations; Black hole binaries

Thesis: Spin mode reconstruction in Lagrangian space

## Fellowships and Awards

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| **Chu Excellence in Research Graduate Student Award**, UIUC | Apr 2025 |
| **Lewis E. Snyder Graduate Student Travel Award**, UIUC | Mar 2025 |
| **CAPS Graduate Fellowship**, Center for AstroPhysical Surveys, UIUC | 2022–2023 |
| **Outstanding Undergraduate Student Award**, XMU | Jun 2021 |
| **Caiwenzhong Fellowship**, College of Physics Science and Technology, XMU | Apr 2020 |
| **Academic Excellence Scholarship**, Department of Astronomy, XMU | Oct 2019 |
| **Guangqi Fellowship**, Shanghai Astronomical Observatory, CAS | May 2019 |
| **National Scholarship**, PRC Ministry of Education | Nov 2018 |

## Publications

NASA ADS records (Oct 2026): total citations 600+ (h-index 9).

**First-author and major-contribution papers**

- **Wu, Q.**, & Shen, Y. (2026). *A Catalog of Spectral Properties for DESI DR1 Quasars.* arXiv preprint arXiv:2610.31416.
- **Wu, Q.**, Shen, Y., Done, C., Goad, M. R., & Hagen, S. (2026). *Understanding the Broad-line Region of Active Galactic Nuclei with Photoionization. II. Slim Disks, Self-shadowing, and BLR Sizes.* arXiv preprint arXiv:2606.00992.
- **Wu, Q.**, Shen, Y., Guo, H., et al. (2025). *Understanding the Broad-line Region of Active Galactic Nuclei with Photoionization. I. The Moderate-Accretion Regime.* The Astrophysical Journal, 980(1), 134.
- **Wu, Q.**, & Shen, Y. (2023). *Improved Redshifts for DESI EDR Quasars.* Research Notes of the AAS, 7(9), 190.

· **Wu, Q.**, & Shen, Y. (2022). *A Catalog of Quasar Properties from Sloan Digital Sky Survey Data Release 16*. The Astrophysical Journal Supplement Series, 263(2), 42.

· Zheng, W. M., **Wu, Q.**, Wu, J., et al. (2022). *The Disk Veiling Effect of the Black Hole Low-mass X-Ray Binary A0620-00**. The Astrophysical Journal, 925(1), 83.

· **Wu, Q.**, Yu, H. R., Liao, S., & Du, M. (2021). *Spin Mode Reconstruction in Lagrangian Space*. Physical Review D, 103(6), 063522.

**Other contributing-author papers**

· Buchner, J., et al. (2026). *A large population of over-massive black hole quasars at $z = 0.3$–$0.8$ revealed by eROSITA*. arXiv preprint arXiv:2603.22425.

· SDSS Collaboration, et al. (2026). *The Nineteenth Data Release of the Sloan Digital Sky Survey*. The Astrophysical Journal Supplement Series, 285(1), 9.

· Aydar, C., et al. (2026). *Robust active galactic nucleus and host-galaxy decomposition in optical spectral fitting*. Astronomy & Astrophysics, 710, A141.

· Zhao, X., et al. (2026). *Transient Relativistic Iron Emission Line from an X-Ray Flaring Supermassive Black Hole*. The Astrophysical Journal, 1000(1), 28.

· Zhuang, M., et al. (2026). *NEXUS: A Spectroscopic Census of Broad-line AGNs and Little Red Dots at $3 \lesssim z \lesssim 6$*. The Astrophysical Journal, 999(1), 31.

· Kollmeier, J. A., et al. (2026). *Sloan Digital Sky Survey. V. Pioneering Panoptic Spectroscopy*. The Astronomical Journal, 171(1), 52.

· Zhao, X., et al. (2025). *Fast X-Ray Variability from the Coronae of Supermassive Black Holes*. The Astrophysical Journal Letters, 995(2), L72.

· Cheng, X., et al. (2025). *Quest for a Coherent Definition of Weak-line Quasars and Its Physical Implications*. The Astrophysical Journal, 994(2), 213.

· Nandra, K., et al. (2025). *The eROSITA Final Equatorial Depth Survey (eFEDS): The hard X-ray selected sample*. Astronomy & Astrophysics, 693, A212.

· Waddell, S. G. H., et al. (2024). *The eROSITA Final Equatorial Depth Survey (eFEDS): Complex absorption and soft excesses in hard X-ray-selected active galactic nuclei*. Astronomy & Astrophysics, 690, A132.

· Musiimenta, B., et al. (2023). *A new discovery space opened by eROSITA: Ionised AGN outflows from X-ray selected samples*. Astronomy & Astrophysics, 679, A84.

· Cheng, S., et al. (2020). *CUBE: Towards an Optimal Scaling of Cosmological N-body Simulations*. In 2020 20th IEEE/ACM International Symposium on Cluster, Cloud and Internet Computing (CCGRID) (pp. 685–690). IEEE.

· Wen, J., et al. (2019). *GRID: a student project to monitor the transient gamma-ray sky in the multi-messenger astronomy era*. Experimental Astronomy, 48, 77–95.

## Approved Proposals

**Wu, Q. (PI),** Shen, Y. *A Local UV Benchmark for Interpreting Unobscured Coronal-Line AGNs.* 19 orbits, **HST** Cycle 34, HST-GO-18341.

**Wu, Q. (PI),** Shen, Y. *HST UV Spectroscopy of High-accretion-rate AGNs and the Origin of the Offset in the Broad-Line Region Size-Luminosity Relation.* 19 orbits, **HST** Cycle 31, HST-GO-17433.

**Wu, Q. (PI),** Wu, J., Sai, H. *Long-Term Optical Monitoring on the Black Hole Binary MAXI J1820+070.* 120 hours, **LCOGT** 2021 CTAP, CTAP2021-A0019.

## Talks

**Invited talks**

- **WHU AGN/Galaxy Seminar**, "Linking AGN Accretion Flows and Broad-Line Region Properties through Photoionization Modeling", Wuhan University Jul 2026
- **XMU Black Hole Seminar**, "Linking AGN Accretion Flows and Broad-Line Region Properties through Photoionization Modeling", Xiamen University Jul 2026
- **UMich Galaxies Seminar**, "Understanding the AGN Line-Emitting Regions: From Optical Spectral Statistics to Broad-Line Region Photoionization Modeling", University of Michigan Feb 2026
- **UIUC Journal Club**, "Understanding the Broad-Line Region of AGN with Photoionization", UIUC Apr 2025
- **Center for AstroPhysical Surveys Seminar**, "Photoionization in the Quasar Broad-Line Region", UIUC Aug 2023
- **UIUC Journal Club**, "Quasar Spectral Decompositions from Public Surveys", UIUC May 2023
- **AGN Demographics Working Group**, "Tutorial on PyQSOFit", MPE (remote) Nov 2022

**Contributed talks**

- **SDSS-V 2025 Collaboration Meeting**, "From Quasar Spectra to Structure: SDSS-V Quasar Spectral Analysis and AGN Photoionization Studies", MPIA (remote) Jun 2025
- **AGN Surveys Workshop**, "Understanding the Broad-Line Region of AGN with Photoionization", Munich Institute for Astro-, Particle and BioPhysics Mar 2025
- **SDSS-V 2024 Collaboration Meeting**, "Quasar Value-Added Catalog and Visual Inspection", New Mexico State University Jun 2024
- **SDSS-V 2023 Collaboration Meeting**, "AGN Demographics", Simons Foundation, New York Aug 2023
- **The 23rd Guoshoujing Symposium**, "Correlations between Halo Spins and Primordial Perturbations", Zhejiang University May 2021
- **Binary System Seminar**, "Dynamical Measurements of the Black Hole Binary MAXI J1820+070", Nanjing University Nov 2020

## Outreach

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| Mentor, UIUC Society for Equity in Astronomy mentoring program | Fall 2022 – Spring 2024 |
| Volunteer, UIUC girls' astronomy summer camp | Summer 2022 |
| President, Xiamen University Astronomical Club | 2018–2019 |
| Video editor and translator, Mufu Astronomy Team | 2018–2020 |